

Inverse hyperbolic trig functions

Basic skills

Q1. $\cosh x = 3$

$$\frac{e^x + e^{-x}}{2} = 3 \Rightarrow e^x + e^{-x} = 6$$

$$\Rightarrow e^x - 6 + e^{-x} = 0 \Rightarrow e^{2x} - 6e^x + 1 = 0$$

$$\text{so } e^x = \frac{6 \pm \sqrt{6^2 - 4 \times 1 \times 1}}{2} = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$

$$\text{so } x = \ln(3 \pm 2\sqrt{2}) \quad \text{arcsinh } x = \ln(x + \sqrt{x^2 + 1})$$

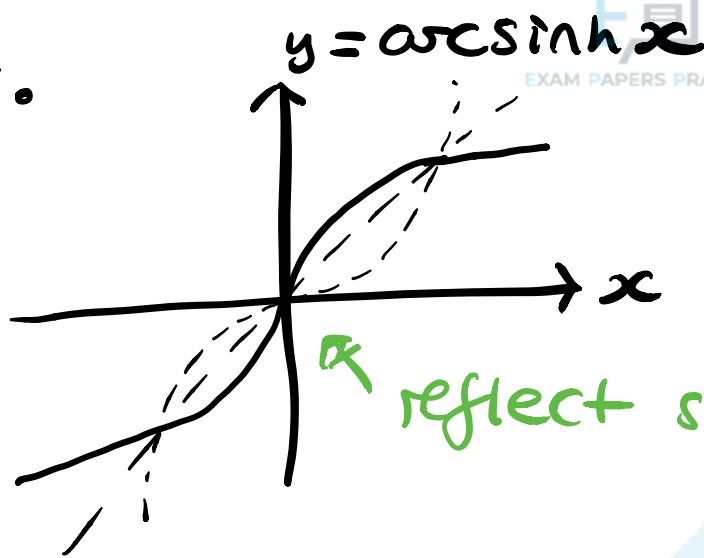
Q2. a) $\operatorname{arcsinh}(2) = \ln(2 + \sqrt{2^2 + 1})$
 $= \ln(2 + \sqrt{5})$

b) $\operatorname{artanh}(-\frac{1}{2}) = \frac{1}{2} \ln\left(\frac{1 + -\frac{1}{2}}{1 - -\frac{1}{2}}\right)$ $\operatorname{artanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$
 $= \frac{1}{2} \ln\left(\frac{\frac{1}{2}}{\frac{3}{2}}\right) = \frac{1}{2} \ln\left(\frac{1}{3}\right)$

Q3. no sol to $\tanh x = 3$
as $-1 < \tanh x < 1$

Q4.

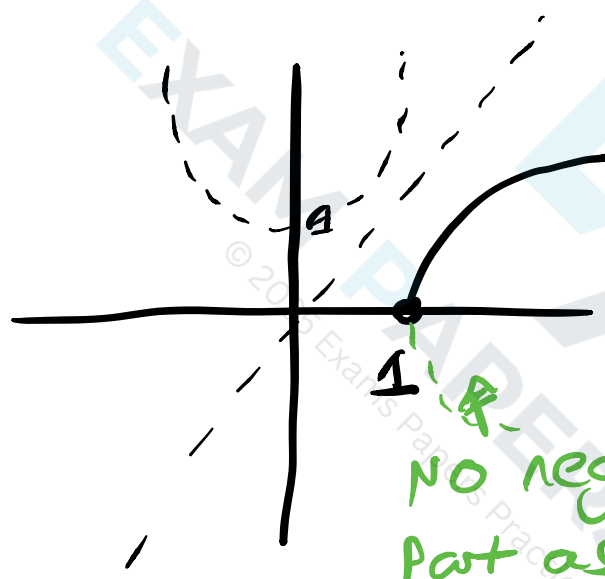
a)



domain: $x \in \mathbb{R}$
 range: $y \in \mathbb{R}$

reflect $\sinh x$ in $y=x$

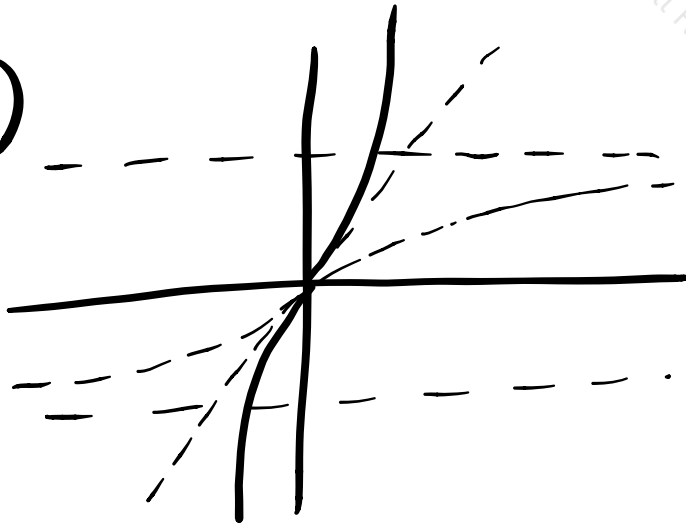
b)



domain: $x \geq 1$
 range: $y \in [0, \infty)$
 $y \geq 0$

NO negative part as function

c)



domain: $-1 < x < 1$
 range: $y \in \mathbb{R}$

Competency skills

Q5. a) $\sinh x = -4 \Rightarrow x = \operatorname{arcsinh}(-4)$

so $x = \ln(-4 + \sqrt{4^2 + 1}) = \ln(\sqrt{17} - 4)$

b) $\cosh x = 3 \Rightarrow x = \operatorname{arcosh}(3)$

so $x = \ln(3 + \sqrt{3^2 - 1}) = \ln(3 + 2\sqrt{2})$

Same as in Q1.

c) $\tanh x = -0.1 \Rightarrow x = \operatorname{artanh}(-0.1)$

so $x = \frac{1}{2} \ln\left(\frac{1 + -0.1}{1 - -0.1}\right) = \frac{1}{2} \ln\left(\frac{0.9}{1.1}\right) = \frac{1}{2} \ln\left(\frac{9}{11}\right)$

$= \frac{1}{2}(\ln 9 - \ln 11) = \ln 3 - \frac{1}{2} \ln 11$

Q6.

a) Let $x = \sinh y$ then $y = \operatorname{arcsinh} x$

so $x = \frac{e^y - e^{-y}}{2} \Rightarrow e^{2y} - 2xe^y - 1 = 0$

so $e^y = \frac{2x \pm \sqrt{(2x)^2 - 4 \times 1 \times -1}}{2} = x \pm \sqrt{x^2 + 1}$

as $x - \sqrt{x^2 + 1} < 0$

only $e^y = x + \sqrt{x^2 + 1}$

so $y = \ln(x + \sqrt{x^2 + 1})$

b) Let $x = \cosh y$ then $y = \operatorname{arccosh} x$

$$\text{So } x = \frac{e^y + e^{-y}}{2} \Rightarrow e^{2y} - 2xe^y + 1 = 0$$

$$\Rightarrow e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$$

both positive so

$$y = \ln(x \pm \sqrt{x^2 - 1})$$

c) Let $x = \tanh y$ so $y = \operatorname{artanh} x$

$$\text{Then } x = \frac{e^y - e^{-y}}{e^y + e^{-y}} \Rightarrow xe^y + xe^{-y} = e^y - e^{-y}$$

$$\Rightarrow (x-1)e^y + (x+1)e^{-y} = 0$$

$$\Rightarrow (x-1)e^{2y} + (x+1) = 0 \Rightarrow e^{2y} = -\frac{x+1}{x-1}$$

$$\text{So } e^{2y} = \frac{1+x}{1-x} \quad \text{so } 2y = \ln\left(\frac{1+x}{1-x}\right)$$

$$\text{Therefore } y = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$$

Q7. $\sinh(2x-1) = 5$

$$\Rightarrow 2x-1 = \operatorname{arcsinh} 5$$

$$\text{so } x = \frac{1}{2} + \frac{1}{2} \operatorname{arcsinh} 5$$

$$= \frac{1}{2} + \frac{1}{2} \ln(5 + \sqrt{5^2 + 1})$$

$$= \frac{1}{2} + \frac{1}{2} \ln(5 + \sqrt{26})$$

Q8. $\cosh^2 x = 4$

$$\Rightarrow \cosh x = \pm 2$$

but $\cosh x \geq 1$ so only $\cosh x = 2$

$$\text{so } x = \operatorname{arcosh} 2$$

$$\Rightarrow x = \ln(2 \pm \sqrt{(2)^2 - 1})$$

$$\text{so } x = \ln(2 \pm \sqrt{3})$$

Advanced skills

Q9. To show $\operatorname{arcosh} x < \operatorname{arsinh} x$

equivalently $\ln(x \pm \sqrt{x^2 - 1}) < \ln(x + \sqrt{x^2 + 1})$

compare

So equivalently

$$x \pm \sqrt{x^2 - 1} < x + \sqrt{x^2 + 1}$$

$$\Leftrightarrow \pm \sqrt{x^2 - 1} < \sqrt{x^2 + 1}$$

$$\Leftrightarrow -1 < 1$$

which is true so hence always

Q10. $x = \sinh(\operatorname{arccosh} 4x)$

$$\Rightarrow \operatorname{arcsinh} x = \operatorname{arccosh} 4x$$

$$\Rightarrow \ln(x + \sqrt{x^2 + 1}) = \ln(4x \pm \sqrt{(4x)^2 - 1})$$

$$\Rightarrow x + \sqrt{x^2 + 1} = 4x \pm \sqrt{16x^2 - 1}$$

$$\text{So } 3x \pm \sqrt{16x^2 - 1} = \sqrt{x^2 + 1}$$

$$(3x \pm \sqrt{16x^2 - 1})^2 = x^2 + 1$$

$$9x^2 \pm 3x\sqrt{16x^2 - 1} + 16x^2 - 1 = x^2 + 1$$

$$\Rightarrow 24x^2 \pm 3x\sqrt{16x^2 - 1} - 2 = 0$$

$$\Rightarrow (24x^2 - 2)^2 = (3x\sqrt{16x^2 - 1})^2$$

$$\Rightarrow 576x^4 - 96x^2 + 4 = 9x^2(16x^2 - 1)$$

$$\Rightarrow 432x^4 - 105x^2 + 4 = 0$$

SO

$$x^2 = \frac{105 \pm \sqrt{105^2 - 4 \times 432 \times 4}}{2 \times 432} \approx 0.196, 0.047$$

SO $x = 0.442, 0.217$

Q11. $\sinh^2 x = 3 (\cosh x - 1)$

↓ as $\cosh^2 x - \sinh^2 x = 1$

$$\cosh^2 x - 1 = 3 \cosh x - 3$$

$$c^2 - 3c + 2 = 0 \quad c = \cosh x$$

$$(c-1)(c-2) = 0 \quad \text{SO } c = 1, 2$$

$$\cosh x = 1 \quad \text{or} \quad \cosh x = 2$$

$x = 0$  $\text{or} \quad x = \operatorname{arccosh} 2$
 $= \ln(2 \pm \sqrt{2^2 - 1})$
 $= \ln(2 \pm \sqrt{3})$

Point on graph for y-int

Q12.

a) $2 \cosh x - 3 \sinh x = 0$
 $\Rightarrow 3 \sinh x = 2 \cosh x$

$$\Rightarrow \tanh x = \frac{2}{3}$$

$$\text{so } x = \operatorname{arctanh} \frac{2}{3} = \frac{1}{2} \ln \left(\frac{\frac{2}{3}}{\frac{1}{3}} \right) = \frac{1}{2} \ln 5$$

b)

$$a \times \frac{e^x + e^{-x}}{2} - b \times \frac{e^x - e^{-x}}{2} = c$$

$$\Rightarrow ae^x + ae^{-x} - be^x + be^{-x} = 2c$$

$$\Rightarrow (a-b)e^{2x} - 2ce^x + (a+b) = 0$$

$$\text{For } e^x = \frac{2c \pm \sqrt{(2c)^2 - 4(a-b)(a+b)}}{2(a-b)}$$

$$= \frac{2c \pm \sqrt{4c^2 - 4a^2 + 4b^2}}{2(a-b)}$$

$$= \frac{c \pm \sqrt{c^2 + b^2 - a^2}}{(a-b)}$$

so for exactly one $c^2 + b^2 - a^2 = 0$
 and need $\frac{c}{a-b} > 0$ so if $c > 0, a > b$
 and $c^2 = a^2 - b^2$
 or if $c < 0, a < b$ and $c^2 = a^2 - b^2$